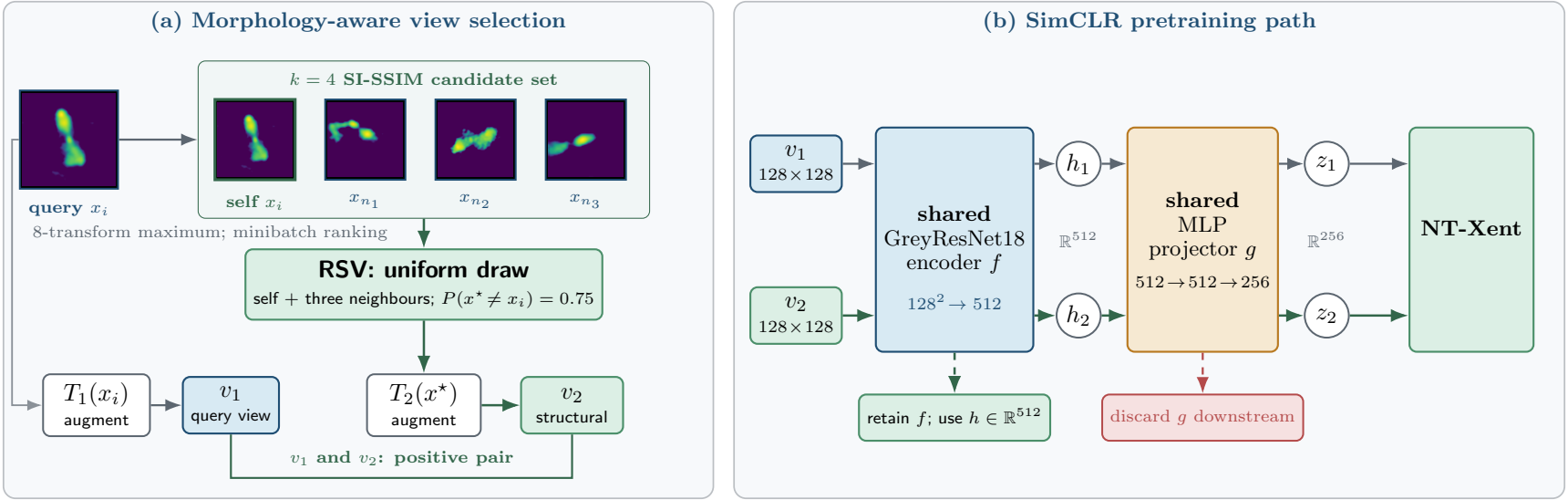

(a) Morphology-aware view selection
k = 4 SI-SSIM candidate set
query x_i
self x_i
x_{n_1}
x_{n_2}
x_{n_3}
8-transform maximum; minibatch ranking
RSV: uniform draw
self + three neighbours; $P(x^\star \neq x_i) = 0.75$
$T_1(x_i)$
augment
v_1
query view
$T_2(x^\star)$
augment
v_2
structural
v_1 and v_2: positive pair
(b) SimCLR pretraining path
v_1
128×128
v_2
128×128
shared
GreyResNet18
encoder f
$128^2 \to 512$
h_1
h_2
$\mathbb{R}^{512}$
shared
MLP
projector g
$512 \to 512 \to 256$
z_1
z_2
$\mathbb{R}^{256}$
NT-Xent
retain f; use $h \in \mathbb{R}^{512}$
discard g downstream